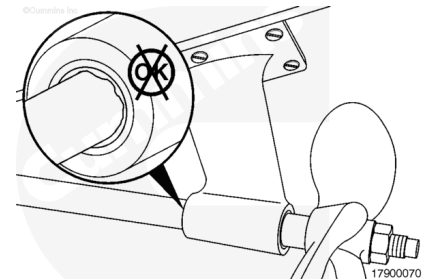


Trainings ▾

General Information

Improper alignment of the propeller shaft can result in many problems. Vibration is usually the first indicator. Failure of the shaft seal or stuffing box can also occur. If damaged shaft seals, stuffing box, strut, cutlass bearing, or propeller are found, refer to an authorized original equipment manufacturer (OEM) repair location.



Preparatory Steps

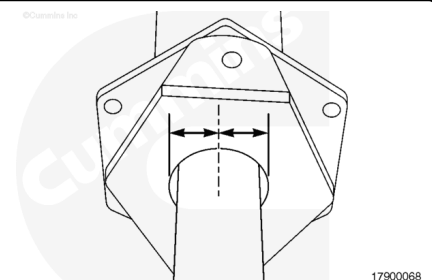
Shaft alignment is checked when isolators are replaced, adjusted, or any time excessive vibration has been noticed. Also, any time the vessel has been taken out of the water and stored or blocked minor changes can take place in the shape of the hull. The shaft is checked and realigned as necessary after the vessel has been placed back in the water.



Cleaning Machine Operation

Inspect the propeller shaft for debris and burrs. Clean the propeller shaft.

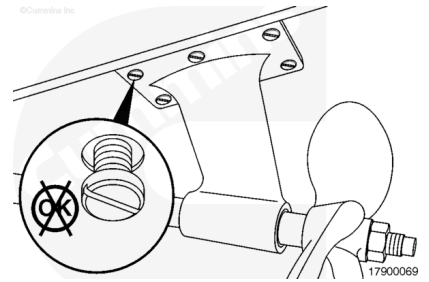
Check that the shaft is centered in the opening of the stuffing box flange. The specifications will vary by manufacturer. If any problems are noted, refer to an authorized OEM service location.



Inspect the strut. Make sure the strut is mounted solidly to the hull.

Check that the shaft is aligned with the cutlass bearing, and that the bearing is **not** worn.

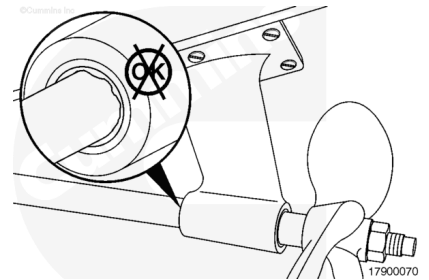
If any problems are noted, refer to an authorized OEM service location.



A misaligned shaft will cause uneven wear of the cutlass bearing.

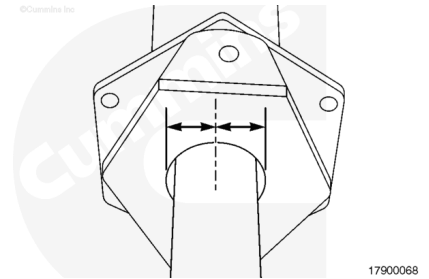
If the strut or cutlass bearing needs to be repaired or replaced, refer to an authorized OEM service location.

Note : Initially align the shaft out of the water using the following instructions, then perform a final alignment after the vessel is in the water.



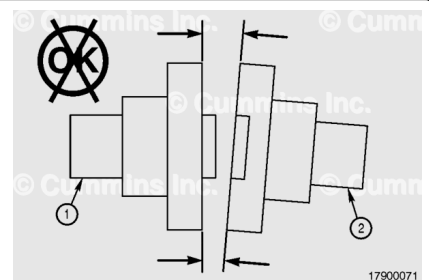
Alignment:

Check that the shaft is centered in the opening of the stuffing box flange.



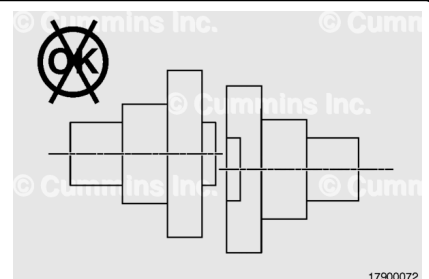
Unbolt the propeller shaft coupling from the marine gear coupling.

The faces of the marine gear (1) couplings and propeller (2) **must** be parallel.

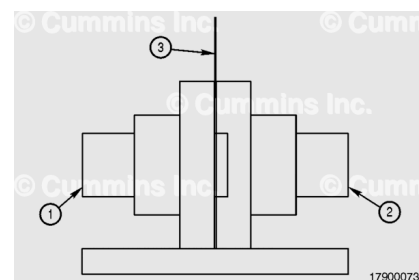


The couplings **must** be aligned.

A straight edge can be used to help align the couplings.



Use a feeler gauge (3) to check the alignment of the coupling in four different places, 90 degrees apart. The variation is to be within 0.051 to 0.102 mm [0.002 to 0.004 in].



If the flange is out of alignment the engine mounting will need to be adjusted until the alignment is correct. Refer to Engine Mounting/Drive Systems section in the Marine Recreational Installation Directions, Bulletin 3884649.

Make sure all mounting bolts and nuts are torqued to the proper specifications. See the manufacturer service information.

Bolt the coupling in place and torque to the specifications for the capscrew and bolt size.

Specifications

Capscrews, Bolts and Nuts Torque for Fine and Coarse Threads				
	SAE Grade 5		SAE Grade 8	
	(1) As Received	(2) Lubricated	(1) As Received	(2) Lubricated
	N•m [ft-lb]	N•m [ft-lb]	N•m [ft-lb]	N•m [ft-lb]
1/4	12 ± 1 [9 ± 1]	9 ± 1 [7 ± 1]	19 ± 1 [14 ± 1]	15 ± 1 [11 ± 1]
5/16	26 ± 3 [19 ± 2]	20 ± 3 [15 ± 2]	37 ± 3 [27 ± 2]	30 ± 3 [22 ± 2]
3/8	45 ± 4 [33 ± 3]	37 ± 3 [27 ± 2]	62 ± 5 [46 ± 4]	52 ± 4 [38 ± 3]
7/16	71 ± 5 [52 ± 4]	54 ± 4 [40 ± 3]	99 ± 8 [73 ± 6]	81 ± 7 [60 ± 5]
1/2	108 ± 8 [80 ± 6]	88 ± 7 [65 ± 5]	152 ± 11 [112 ± 8]	122 ± 9 [90 ± 7]
9/16	152 ± 11 [112 ± 8]	112 ± 11 [90 ± 8]	214 ± 16 [158 ± 12]	176 ± 14 [130 ± 10]
5/8	214 ± 16 [158 ± 12]	176 ± 14 [130 ± 10]	304 ± 22 [224 ± 16]	244 ± 20 [180 ± 15]
3/4	380 ± 27 [280 ± 20]	305 ± 27 [225 ± 20]	529 ± 41 [390 ± 30]	434 ± 34 [320 ± 25]
7/8	607 ± 43 [448 ± 32]	488 ± 41 [360 ± 30]	854 ± 68 [630 ± 50]	691 ± 54 [510 ± 40]
1	922 ± 68 [680 ± 50]	732 ± 61 [540 ± 45]	1302 ± 95 [960 ± 70]	1051 ± 81 [775 ± 60]
1 1/8	1152 ± 81 [850 ± 60]	915 ± 81 [675 ± 60]	1844 ± 136 [1360 ± 100]	1491 ± 115 [1100 ± 85]

